

Machine Learning — Introduction & Complete Workflow

Beginner-friendly notes based primarily on the uploaded Machine Learning (2).pdf

1. What is Machine Learning?

Machine Learning is a way of teaching computers to learn patterns from data and use those learned patterns to make predictions or decisions. Instead of manually programming every rule, we provide examples/data and allow an algorithm to learn a model.

Simple idea: Data + learning algorithm → trained model → prediction/decision

Traditional Programming	Machine Learning
Rules + data → result	Data + known results → learned model
Logic is manually written	Patterns are learned from examples
Rules are explicit	Useful when rules are difficult to hand-code

2. Real-World Examples

- Recommendation systems — learn from user/watch history.
- Spam detection — learn patterns associated with spam emails.
- Voice assistants — learn speech patterns.
- Face recognition — learn visual patterns.
- Self-driving systems — identify roads, signs and pedestrians.

3. AI vs ML vs Deep Learning

AI is the broad field of building systems that perform tasks associated with intelligence. **ML** is a subset of AI where patterns are learned from data. **DL** is a subset of ML that uses neural networks, particularly useful for complex data such as images, speech and language.

AI	ML	DL
Broad umbrella	Subset of AI	Subset of ML
Can use many approaches	Learns patterns from data	Uses multi-layer neural networks
Chess, autonomous systems	Spam, recommendations	Vision, speech, large neural models

4. Types of Machine Learning

Supervised Learning: training data contains inputs and known outputs/labels. Example: income + credit score → loan approval. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, Naive Bayes, KNN, SVM and Gradient Boosting.

Unsupervised Learning: there is no target label; the objective is to discover patterns, groups, structure or anomalies.

Reinforcement Learning: an agent learns through interaction with an environment and feedback in the form of rewards/penalties.

5. Complete Machine Learning Workflow

The uploaded PDF gives an 11-step workflow. The first five steps are especially important for data understanding and preparation.

#	Step	What happens?
1	Problem definition	Define the prediction/task, target and success criteria.
2	Data collection	Collect relevant data from files, databases, APIs, sensors, etc.
3	EDA	Understand structure, distributions, missingness, relationships and anomalies.
4	Data preprocessing / cleaning	Fix data-quality problems and transform data into a usable form.
5	Feature selection & engineering	Create useful signals and select relevant features.
6	Split dataset	Create training/test (and possibly validation) sets.
7	Model selection	Choose an algorithm suitable for the task and data.
8	Model training	Fit the model using training data.
9	Model evaluation	Measure performance with suitable metrics.
10	Hyperparameter tuning	Search for useful model settings.
11	Testing / validation	Confirm performance on held-out data and validate the workflow.

6. EDA — Exploratory Data Analysis

EDA is the process of analyzing, visualizing and understanding data before building a machine-learning model. It is detective work: understand the dataset, discover patterns, identify anomalies, generate insights and decide what to do next.

EDA checklist:

- View the data: `head()`, `tail()`, `shape`, `info()`, columns and data types.
- Summary statistics: mean, median, mode, standard deviation, min, max and quartiles.
- Value counts for categorical columns.
- Missing-value analysis.
- Visualizations: histograms, boxplots, bar plots, scatter plots and correlation heatmaps.
- Target-variable exploration for supervised learning.

7. Data Cleaning

Cleaning focuses on fixing data-quality problems identified during exploration.

- Handle missing values: drop only when justified; otherwise use appropriate imputation.
- Remove duplicate rows.
- Fix wrong data types.
- Standardize inconsistent categories such as Male/male/MALE.
- Detect and handle outliers using approaches such as IQR, boxplots or z-scores.
- Check domain/logic errors such as impossible ages or implausible measurements.

8. Data Preprocessing

Preprocessing transforms valid/clean data into a representation suitable for analysis and ML models.

- Encoding categorical variables: ordinal/label encoding for ordered categories; one-hot encoding for nominal categories.
- Feature transformations such as log or square-root transformations for skewed data.
- Feature scaling: Min-Max normalization or Standardization (Z-score).
- Feature engineering: mathematical combinations, flags, binning and time-based features.
- Feature selection: filter methods such as correlation/chi-square/ANOVA and embedded methods such as Lasso or tree-based importance.

9. Key Rule — Do Not Leak Test Information

When preprocessing for a predictive model, split the data before learning preprocessing parameters. Fit imputers, scalers and encoders on the training data, then transform the test data using those fitted objects. This prevents information from the test set influencing training.

Source mapping: the uploaded 60-page PDF covers the introduction and ML types in approximately pages 2–28, the complete workflow on page 30, EDA on pages 29–39, cleaning on pages 40–47, preprocessing on pages 48–55, and feature engineering/selection on pages 56–60.